Grounded Headline Generation from Public Event Images

LLM Programming Task 20: Headline Generation

1. Title and Group Details

Project title: Grounded Headline Generation from Public Event Images

Name	Roll No.	Contribution
Aditya Kumar	2330133	Image captioning (BLIP) and event-type detection (CLIP). Collected the test images and ran the batch tests on Colab.
Aman Kumar	2330065	Headline generation with Qwen2.5 (prompt design), OCR stage and its noise filter, and the grounding check with retry and fallback.
Aman Kumar	2330361	Gradio demo app, screenshots, GitHub repository, report, slides and the presentation video.

2. Problem Statement and Objectives

Problem. Given a photograph of a public event such as a rally, march, festival, concert or sports match, generate a short, descriptive news-style headline.

Motivation. Newsrooms and photo archives need quick headlines for large collections of images. Language models tend to invent names, places, dates and causes that a photograph cannot support, which is unacceptable in journalism. The aim is a headline that stays within what the image actually shows.

Objectives.

1. Generate fluent, short headlines (about 6 to 14 words) from a single image.
2. Restrict the content to what is visible: the scene, the activity and any legible text.
3. Automatically detect and suppress unsupported names and numbers.

Expected outcome. A working prototype that returns a headline together with the evidence it was built from (caption, event type, visible text, grounding status).

3. Proposed Methodology and Processing Pipeline

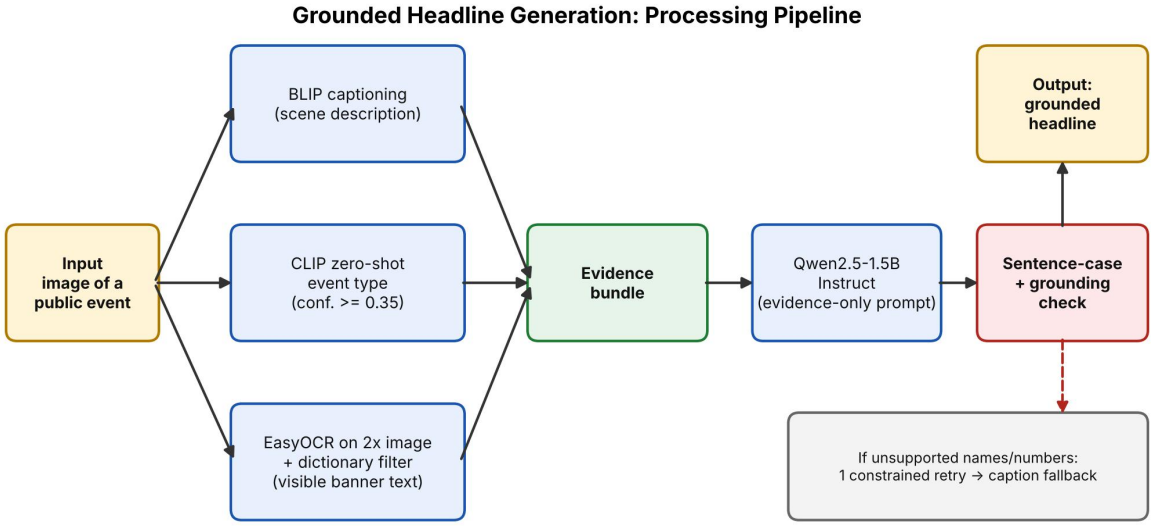


Figure 1: Processing pipeline

Input. One RGB image of a public event.

Intermediate processing.

1. Captioning: BLIP produces a one-sentence description of the scene.
2. Event typing: CLIP scores the image against 15 generic event labels (political rally, protest march, music concert, running race, press conference and so on). The label is used only if its probability is at least 0.35.
3. OCR: EasyOCR reads text on a 2x upscaled copy of the image. Only detections with confidence of at least 0.4 are kept, and only alphabetic dictionary words of three or more letters, which removes OCR noise.
4. Evidence bundle: the caption, event type and visible text are combined into a short block of text.
5. Headline writing: Qwen2.5-1.5B-Instruct is prompted to use only the evidence, with greedy decoding.
6. Formatting: the output is converted to sentence case. Words stay capitalised only if they appear capitalised in the OCR text.
7. Grounding check: every number and every capitalised non-initial word in the headline must appear in the evidence. If not, the model is retried once with the unsupported words banned in the prompt. If the retry also fails, the system falls back to a cleaned version of the BLIP caption.

Output. The headline, a flag showing whether the fallback was used, and the supporting evidence.

4. Models and Tools Used

Component	Model / library	Source
Captioning	Salesforce/blip-image-captioning-large	https://huggingface.co/Salesforce/blip-image-captioning-large
Event typing	openai/clip-vit-base-patch32	https://huggingface.co/openai/clip-vit-base-patch32
Headline LLM	Qwen/Qwen2.5-1.5B-Instruct	https://huggingface.co/Qwen/Qwen2.5-1.5B-

Component	Model / library	Source
		Instruct
OCR	EasyOCR	https://github.com/JaidedAI/EasyOCR
OCR filtering	pyspellchecker	https://pypi.org/project/pyspellchecker/
Demo UI	Gradio	https://www.gradio.app

Frameworks and environment: Python 3, PyTorch, Hugging Face Transformers, Pillow, NumPy. Development and testing were done on Google Colab with a T4 GPU. All models are used as released, with no training or fine-tuning, so no training dataset was needed.

5. Implementation and Usability

Implementation. The pipeline is a single class, HeadlineGenerator, in headline_generator.py. Models load lazily on first use. A generate() call returns a result object holding the caption, event type and confidence, OCR text, headline, a grounded flag and a fallback flag. app.py wraps this in a Gradio interface that shows the headline next to its evidence, so a human editor can see why a headline was produced.

Running the application.

1. Clone the repository and run pip install -r requirements.txt.
2. Start the demo with python app.py, then upload an image and press Submit.
3. Alternatively call HeadlineGenerator().generate(image) from a notebook. A batch cell that runs every image in samples/ and writes outputs/results.csv is included in the README.
4. On Colab, select a T4 GPU runtime. The first run downloads roughly 5 GB of model weights.

Practical use. Draft headlines for photo desks and event archives, alt-text and caption suggestions, and a first pass for social media posts. The output is meant to be reviewed by a person, and the evidence panel makes that review quick.

6. Results and Sample Outputs

The system was run on 8 test images collected from the web. The table lists the final headline for each and our assessment against the image.

Image	Scene type	Generated headline	Assessment
download.jpg	Street market	A bustling market thrives with shoppers	Mostly. "Bustling" and "thrives" are interpretation, not strictly visible.
images (1).jpg	Protest march	Protesters march for fossil march	No. OCR fragments of the banner were merged into a meaningless phrase.
images (2).jpg	Aerial festival crowd	Aerial view shows religious festival crowd procession	Partly. "Religious" comes from the CLIP guess, not legible text.
images (3).jpg	Concert	Crowd watches band perform at concert	Yes.
images (4).jpg	Sport	Men squatting during squash game	Doubtful. Sport and "squatting" come from the captioner and may be wrong.

Image	Scene type	Generated headline	Assessment
images (5).jpg	Running race	Group of men run marathon race city street	Partly. "Marathon" is a specific claim from a visual guess.
images (6).jpg	Press conference	People sit in chairs during a press conference	Yes. Generic but accurate.
images.jpg	Political rally	Man speaks at political rally before large crowd	Yes.

The grounding fallback was not triggered in the final run of these 8 images, so every headline in the table came from the language model.

Screenshots

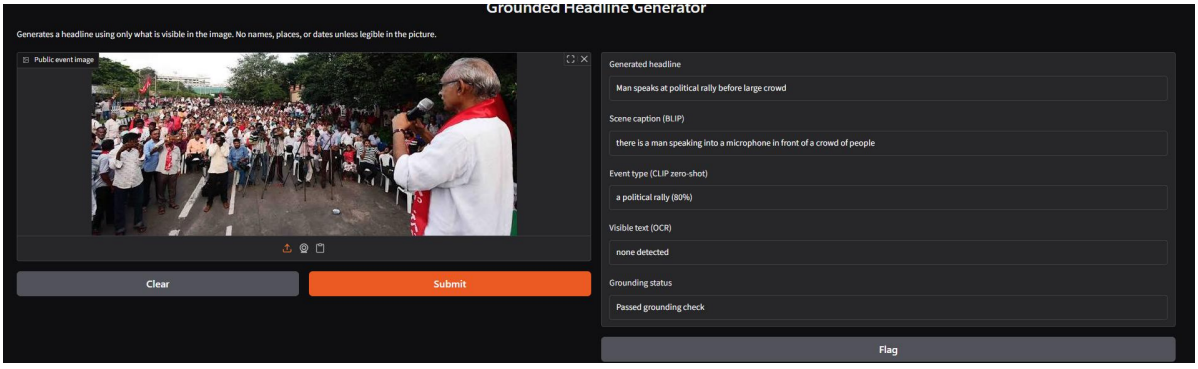


Figure 2: Political rally. Caption and event type are correct and the headline passed the grounding check.

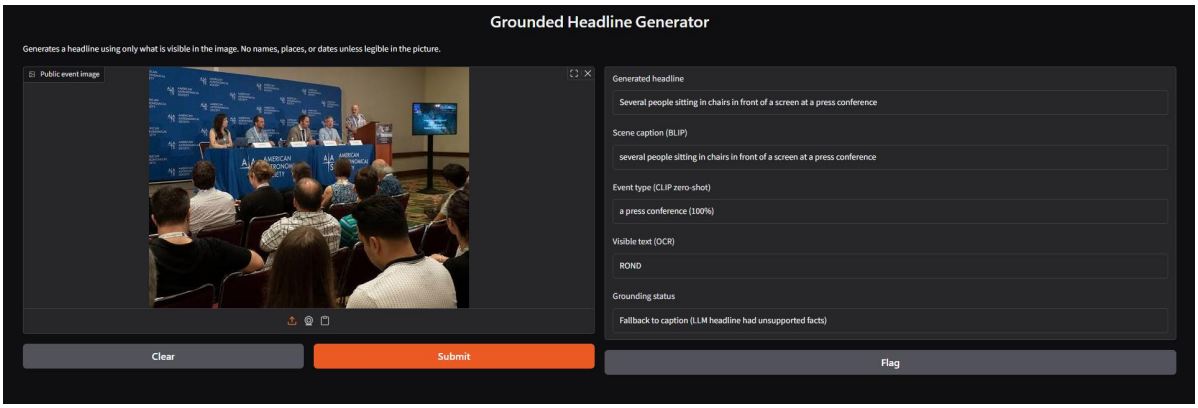


Figure 3: Press conference (an earlier version of the code). OCR returned only noise ("ROND"), the first headline contained unsupported content, and the system fell back to the caption.

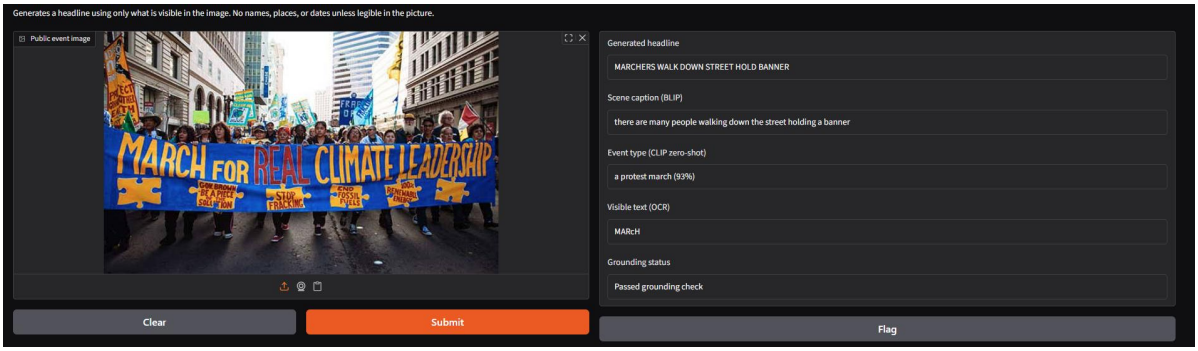


Figure 4: Climate march (an earlier version of the code). The banner is legible but OCR returned only "MARcH", and the model wrote the headline in capital letters.

Improvements made during testing

Observation	Change
Headlines sometimes came out in capitals or Title Case	Added a sentence-case step that keeps capitals only for words capitalised in the OCR text
Fallback captions began with "There is" or "This is"	Strip these prefixes in the fallback
OCR missed small or angled banner text	Upscale the image 2x before OCR and lower the confidence threshold to 0.4
OCR noise such as "GMERICAN" reached the evidence	Keep only dictionary words of three or more letters
"Marathon" was a specific claim from a visual guess	Changed the CLIP label to the more general "a running race"

Observations

- The pipeline produces short, fluent headlines for rallies, concerts, markets, sports and press events, and in the cleanest cases (rally, concert, press conference) the headline matches the image.
- The grounding check did its job on the press conference image: it rejected an unsupported headline and the system fell back to a safe caption.
- No quantitative score was computed. The assessment above is a manual judgement on a small set of 8 images and should not be read as a measured accuracy.

Limitations

- The grounding check is lexical. It compares words, so it cannot tell that a headline is wrong when every word appears in the evidence. This is what happened on the climate march image, where banner fragments were combined into "Protesters march for fossil march".
- It also cannot catch unsupported adjectives or causes, such as "bustling" and "thrives" in the market headline.
- OCR still struggles with large stylised, angled or repeated text, so the best fact in a photo can be missed.
- The captioner can mislabel details such as the sport being played or what people are doing, and the headline inherits the error.
- Event type comes from a zero-shot guess, which can add a specific word such as "religious" that the image does not make explicit.
- The system does not identify people, places or dates by design, so headlines stay general unless text is legible.
- The test set is small (8 images), so the results are illustrative only.

7. Conclusion and Future Scope

We built a working prototype that turns an event photo into a short headline using only evidence extracted from the image, and that rejects headlines containing unsupported names or numbers. Testing showed that it works well on clear scenes and fails safely when the evidence is weak. It also showed the limits of a purely lexical grounding check.

Possible improvements:

- Replace the multi-stage pipeline with a single vision-language model such as Qwen2-VL.

- Add an entailment (NLI) based faithfulness check to catch unsupported adjectives and meaning errors.
- Use a stronger OCR model with text detection tuned for banners and signs.
- Support Hindi and other Indian languages for both OCR and headlines.
- Evaluate on a larger set of images with human ratings for fluency and faithfulness.

8. References and GitHub Repository Link

- Li, J. et al. BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation. 2022.
- Radford, A. et al. Learning Transferable Visual Models From Natural Language Supervision (CLIP). 2021.
- Qwen Team. Qwen2.5 Technical Report. 2024.
- EasyOCR, JaidedAI. https://github.com/JaidedAI/EasyOCR
- Hugging Face Transformers. https://huggingface.co/docs/transformers

GitHub repository: https://github.com/Aditya-kr24/Headline_Generator